

Does M&A *Create Value?*

A long-run performance study of 786 US acquisitions, 2012–2021

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ABSTRACT

This paper follows 786 significant acquisitions completed by US listed non-financial companies between 2012 and 2021, each worth at least 10% of the acquirer's market value and together worth \$1,802.7bn, and asks whether the acquirer's shareholders did better or worse than holders of comparable stocks over the following three to five years. The sample is rebuilt entirely from public sources: deal values from XBRL filings, completion dates from 8-K Item 2.01 reports, prices and dividends including delisted securities, and Fama-French benchmark portfolios, all as available on 31 March 2025.

The average acquirer trails its size and book-to-market benchmark by 12.4% over 36 months (median 21.3%; skewness-adjusted $t = -4.11$), and only 37% beat it. Calendar-time regressions, which handle cross-correlation and time-varying risk, shrink the shortfall to -0.19% a month equal-weighted (significant only at 10%) and -0.10% value-weighted (not statistically significant). The clearest signal is the method of payment: acquirers that paid partly or mostly in stock earned -0.58% a month ($t = -3.18$), against -0.10% for cash acquirers. Industry-adjusted operating returns show no improvement beyond mean reversion, and 47% of acquirers booked a goodwill impairment within five years.

Keywords: mergers and acquisitions, long-run abnormal returns, BHAR, calendar-time portfolios, method of payment, operating performance, goodwill impairment. JEL: G14, G34.

786 acquisitions, 2012–2021, each at least 10% of the acquirer	–12.4% mean 36-month BHAR vs size and book-to-market peers	–6.8% yearly five-factor alpha when the deal was paid partly or mostly in stock	47% of acquirers impaired goodwill within five years
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1 Introduction

Acquisitions are the largest investment decisions most companies make, and they are usually judged in the few days around the announcement. The announcement return is a forecast: it tells us what investors expected the deal to do, given what they knew. Whether the expectation was right is a different question, and it can only be answered years later, once integration has happened, synergies have or have not shown up, and the price paid has been tested against the cash the business produced. This paper asks that second question for a recent decade of US deals.

The question matters for boards and for the bankers who advise them. A board approving a large acquisition is implicitly betting that its company will be worth more with the target than without it, net of the premium paid. If the typical large deal leaves the acquirer behind comparable companies for years afterwards, the starting point of every deal discussion should be that base rate, and the burden of proof should sit with the deal case. If the typical deal does fine, but some kinds of deal do badly, the useful output is a list of warning signs.

The academic literature has studied long-run acquirer performance since the early 1990s and has not settled it. Early studies found large negative abnormal returns for acquirers over three to five years. Later work showed that much of that evidence is sensitive to how abnormal returns are measured, and that long-horizon tests are easy to get wrong. Most of that literature also ends before 2010. The 2012 to 2021 period is different in ways that could matter: interest rates were low for most of it, large growth stocks dominated market returns, and goodwill accounting moved to impairment-only testing well before the start of the sample, so impairments can be tracked directly.

The study makes three choices that shape the results. First, the sample is built only from public, free sources, so every step can be reproduced: the SEC's XBRL frames give each company's cash paid for acquisitions, stock issued for acquisitions and consideration transferred; 8-K Item 2.01 filings give the completion date; a community price database that keeps delisted stocks gives monthly total returns; and the Kenneth French data library gives benchmark portfolios and factors. Second, only deals that matter to the acquirer are kept: at least \$100m and at least 10% of the acquirer's market value, with a median of 27%. Third, every result is shown with the two main long-horizon methods side by side, buy-and-hold abnormal returns (BHAR) against size and book-to-market portfolios, and calendar-time portfolio regressions, because the two answer slightly different questions and disagree more often than one would like.

The findings fit in four sentences. The average acquirer underperforms its benchmark in event time, by 12.4% over three years, and the result survives every sample and benchmark variation tried. In calendar time and after controlling for five factors and momentum, the average shortfall is much smaller and statistically weak, so a good part of the event-time gap reflects the kind of company that acquires rather than the acquisition itself. In calendar time the clearest dividing line is payment: stock-financed acquirers underperform by about 7% a year, while cash acquirers are close to zero; in event time the gap shows up in the median rather than the mean. And the accounts do not show the operating improvement that would justify the premiums paid: industry-adjusted margins fall after the deal no more and no less than mean reversion predicts, and almost half of acquirers write goodwill down within five years.

Section 2 reviews the literature. Section 3 describes the data and sample. Section 4 sets out the methods. Section 5 presents the results, section 6 discusses what they mean for a deal team, and section 7 lists the limits.

2 What the literature says

Early evidence of underperformance. Agrawal, Jaffe and Mandelker (1992) found that NYSE acquirers lost about 10% over the five years after a merger, after adjusting for size and beta. Loughran and Vijh (1997) split acquisitions by type and payment and found that stock mergers were followed by large negative excess returns over five years while cash tender offers were followed by positive ones. Rau and Vermaelen (1998) showed that the underperformance was concentrated in "glamour" acquirers with low book-to-market ratios, and argued that markets and managers both extrapolate past performance.

Why payment should matter. Myers and Majluf (1984) showed that issuing equity signals that managers think it is overvalued. Shleifer and Vishny (2003) built a model in which overvalued firms use their stock to buy less overvalued ones: the deal can be rational for the acquirer's long-term holders even if the stock later falls, because it would have fallen more without the deal. Savor and Lu (2009) found support for that view by using failed stock bids as a counterfactual. The prediction relevant here is simply that stock acquirers should underperform after the deal, whether or not the deal itself destroyed value.

Why measurement is hard. Barber and Lyon (1997) and Kothari and Warner (1997) showed that long-horizon abnormal returns are biased and skewed, that test statistics are badly specified in random samples, and that the choice of benchmark matters. Lyon, Barber and Tsai (1999) recommended reference portfolios on size and book-to-market with a skewness-adjusted, bootstrapped t-statistic, which is the event-time approach used here. Fama (1998) argued that long-term anomalies tend to shrink or disappear with reasonable changes in method, and favoured calendar-time portfolios, which avoid compounding and account for cross-sectional dependence. Mitchell and Stafford (2000) applied both approaches to acquisitions, seasoned equity offerings and repurchases and found that much of the apparent acquirer underperformance does not survive the calendar-time approach, with stock-financed acquirers the main exception. Loughran and Ritter (2000) pointed out the other side: value-weighted calendar-time tests have low power because mispricing is concentrated in smaller firms.

Operating performance and announcement returns. Healy, Palepu and Ruback (1992) studied the 50 largest US mergers of 1979 to 1984 and found that industry-adjusted operating cash flow returns improved after the merger, a result that relies on regressing post-merger on pre-merger performance to strip out mean reversion. Moeller, Schlingemann and Stulz (2005) documented that acquiring-firm shareholders lost around \$240 billion at announcement in 1998 to 2001, concentrated in a small number of very large deals by highly valued acquirers. Andrade, Mitchell and Stafford (2001) summarised the evidence as positive combined announcement returns, near-zero acquirer announcement returns and long-run acquirer returns that are negative but fragile.

This study sits between those strands. It uses the event-time method recommended by Lyon, Barber and Tsai and the calendar-time method favoured by Fama and Mitchell and Stafford on the same sample, extends the evidence to 2012 to 2021, and adds the operating and impairment record from the same filings.

3 Data and sample

3.1 Identifying deals

There is no free, complete database of US acquisitions with completion dates and payment terms. The sample is therefore built bottom-up from what acquirers must file. For every company filing XBRL financial statements with the SEC, the frames API gives annual values of six tags: cash paid to acquire businesses (net of cash acquired, or gross), cash paid for businesses and interests in affiliates, stock issued for acquisitions, consideration transferred, and goodwill acquired. For each company and fiscal year the deal value is the larger of the total paid (cash plus stock), the consideration transferred and, where no payment is tagged, the goodwill acquired. This gives 7,304 company-years with at least \$100m of acquisition spending in fiscal years 2011 to 2022, of which 5,596 are outside finance and utilities.

Spending in the cash-flow statement tells us that an acquisition happened in a fiscal year, not when. The completion date comes from the company's filing history: the first 8-K with Item 2.01 (completion of acquisition or disposition of assets) filed between the start of the fiscal year and ten days after its end. Requiring an Item 2.01 filing also filters out years in which spending was a series of small bolt-ons that do not trigger the filing. 2,007 company-years have one in 2012 to 2021. Month 1 of every return window is the first full calendar month after the filing.

3.2 Prices, dividends and delistings

Monthly total returns come from the post-no-preference/stocks database on DoltHub, which keeps daily prices, splits and dividends for US listed securities from 2011, including stocks that were later delisted. Month-end closes are adjusted for splits, and dividends paid in the month (including special dividends and the value of spin-offs where the database records them) are added back. Matching an SEC filer to a ticker is the most error-prone step, because tickers are reused and company names change. Every candidate ticker is therefore validated: the month-end price times the shares outstanding on the 10-K cover must be within 20% to 120% of the public float the company reported on the same cover. When a firm stops trading before the end of a window it earns its benchmark return for the remaining months, which is the standard treatment and assumes that the delisting value is reflected in the last traded price; 3.9% of acquirers stop trading within 36 months.

Free price data contain errors, and three kinds are corrected, each only when the company's own cover-page share count confirms it: splits missing from the database or recorded where none took place (48 firm-months), a split recorded one month early or late (1), and ticker changes, where the first month under the new symbol is spliced to the last close under the old one (60). Without these repairs a single missing one-for-ten reverse split turns one acquirer's three-year return into +400%.

3.3 Final sample

Table 1 shows how the sample is built. The acquirer's market value at the end of the month before completion is the month-end price times the nearest reported share count. Deals below 10% of that value are dropped, because a deal that small cannot move the acquirer's stock in a measurable way. When an acquirer completes several qualifying deals, only the first in any 36-month period is kept so that event windows of the same firm do not overlap. The final sample has 786 deals, of which 783 have a full 36-month window.

TABLE 1 SAMPLE CONSTRUCTION

STEP	REMAINING
Acquirer-years with \$100m or more spent on acquisitions (XBRL, fiscal 2011 to 2022)	7,304
Excluding banks, insurers and other financials (SIC 6000 to 6999) and utilities (4900 to 4999)	5,596
With an 8-K Item 2.01 (completion of acquisition) in the fiscal year, completed January 2012 to December 2021	2,007
Acquirer matched to a validated monthly price series	1,666
Market value available at the month before completion	1,338
Deal value at least 10% of the acquirer's market value	950
One event per acquirer every 36 months (final sample)	786
With a return in the first month after completion (return tests)	783

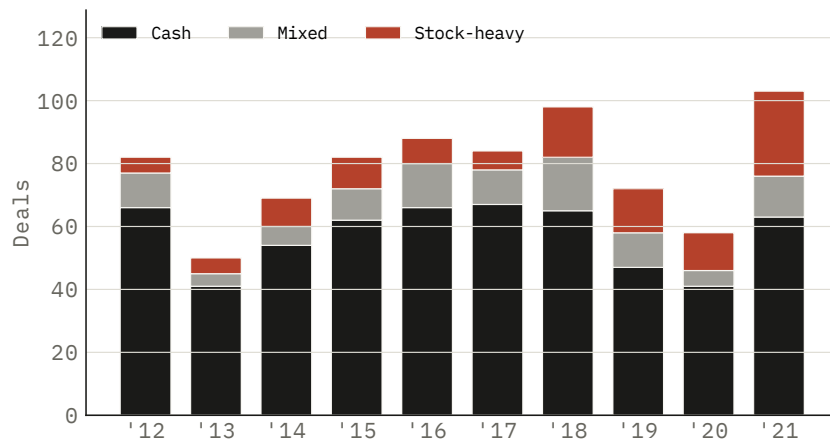
The price match requires the monthly close times the cover-page share count to agree with the public float reported on the 10-K cover (ratio 0.2 to 1.2); 99.7% of final events pass this check, the rest are matched on an identical company name and a ticker that appears in the filings. 3 deals have no return in the first month after completion, a ticker change the price data do not bridge, and drop out of the return tests.

Table 2 describes the sample. The median deal is worth \$618m and 27% of the acquirer's market value, so these are transformational deals for the acquirer rather than routine bolt-ons. 73% were paid essentially in cash according to the cash-flow and equity statements. That share is higher than in most deal databases, partly because the period's low interest rates made debt-financed cash deals cheap and partly because XBRL tagging understates stock consideration in some large mergers, a point returned to in section 7. 36% of acquirers sit in the lowest NYSE book-to-market quintile at completion, so the sample leans towards richly valued companies, as the literature would predict.

TABLE 2 THE SAMPLE AT A GLANCE

	VALUE	
Deals	786	
Aggregate deal value	\$1,802.7bn	
Median deal value	\$618m	
Median acquirer market value before completion	\$2,003m	
Median deal value / acquirer market value (interquartile range)	27%	17% to 48%
Median goodwill recognised / deal value	51%	
Serial acquirers (two or more \$100m+ years in the prior five)	27%	
Shares outstanding up 10% or more around the deal	29%	
Glamour acquirers (lowest NYSE book-to-market quintile)	36%	
Delisted within 36 months (acquired or failed)	3.9%	
PAYMENT, AS DISCLOSED IN CASH-FLOW AND EQUITY STATEMENTS		
Cash	572	73%
Mixed	102	13%
Stock-heavy	112	14%
INDUSTRY (FIRST DIGIT OF SIC CODE)		
Agriculture	2	0%
Mining, oil and gas, construction	79	10%
Manufacturing: food, chemicals, pharma	146	19%
Manufacturing: machinery, electronics, instruments	262	33%
Transport and communications	46	6%
Wholesale and retail	72	9%
Business services and software	135	17%
Health, engineering and other services	44	6%

FIGURE 1 SAMPLE BY COMPLETION YEAR AND DISCLOSED PAYMENT



Deals in the final sample by calendar year of the 8-K Item 2.01 filing.

4 Method

4.1 Buy-and-hold abnormal returns

For deal i and horizon H , the buy-and-hold abnormal return compounds the acquirer's monthly total return and subtracts the compounded return of its benchmark over the same months:

$$BHAR_{i,H} = \prod_{t=1..H} (1 + r_{i,t}) - \prod_{t=1..H} (1 + r_{b(i),t})$$

BHAR measures what an investor who bought the acquirer's stock at completion and held it actually earned relative to an alternative with the same size and valuation profile. The benchmark is one of the 25 Fama-French value-weighted portfolios formed on size and book-to-market, assigned at completion using NYSE breakpoints: size from market value at the month before completion, book-to-market from the latest book equity reported between three and eighteen months earlier. Firms with missing or negative book equity are placed in the middle or lowest book-to-market quintile respectively. Lyon, Barber and Tsai (1999) show that reference portfolios avoid the new-listing, rebalancing and skewness biases of market-index benchmarks, though not all of them.

Because BHARs are strongly right-skewed, the ordinary t-statistic over-rejects for negative means. The paper reports the Johnson skewness-adjusted t-statistic and a p-value from 2,000 bootstrap resamples of it, as well as a Wilcoxon signed-rank test on the median.

4.2 Calendar-time portfolios

Every month from February 2012 to March 2025, a portfolio holds every acquirer that completed a sample deal in the previous 36 months. Its excess return is regressed on the Fama-French (2015) five factors and the Carhart (1997) momentum factor; the CAPM and the Fama-French (1993) three-factor model are shown for comparison:

$$r_{p,t} - r_{f,t} = \alpha + \beta(Mkt - RF) + s \cdot SMB + h \cdot HML + r \cdot RMW + c \cdot CMA + m \cdot MOM + \varepsilon_t$$

The intercept is the average monthly abnormal return. Because each month's portfolio is one observation, cross-correlation between overlapping windows of different acquirers, the main weakness of event-time tests in merger waves, is absorbed into the portfolio's variance. The factor loadings also adjust for risk exposures that shift over time. Standard errors are Newey-West (1987) with three lags. The regression is run with equal and with value weights; the first treats every deal equally, the second answers how much money was lost in aggregate.

4.3 Operating performance and impairments

Following Healy, Palepu and Ruback (1992), operating performance is operating income divided by average total assets, adjusted by subtracting the median of all SEC filers in the same two-digit SIC industry and year. Profitable firms tend to become less profitable over time, so a simple before-and-after comparison would attribute mean reversion to the deal. The regression of the post-deal average (years +1 to +3) on the pre-deal level (year -1) separates the two: the slope captures persistence and the intercept the abnormal change. Goodwill impairments come from the GoodwillImpairmentLoss tag in the deal year and the following five fiscal years.

4.4 Cross-section and placebo

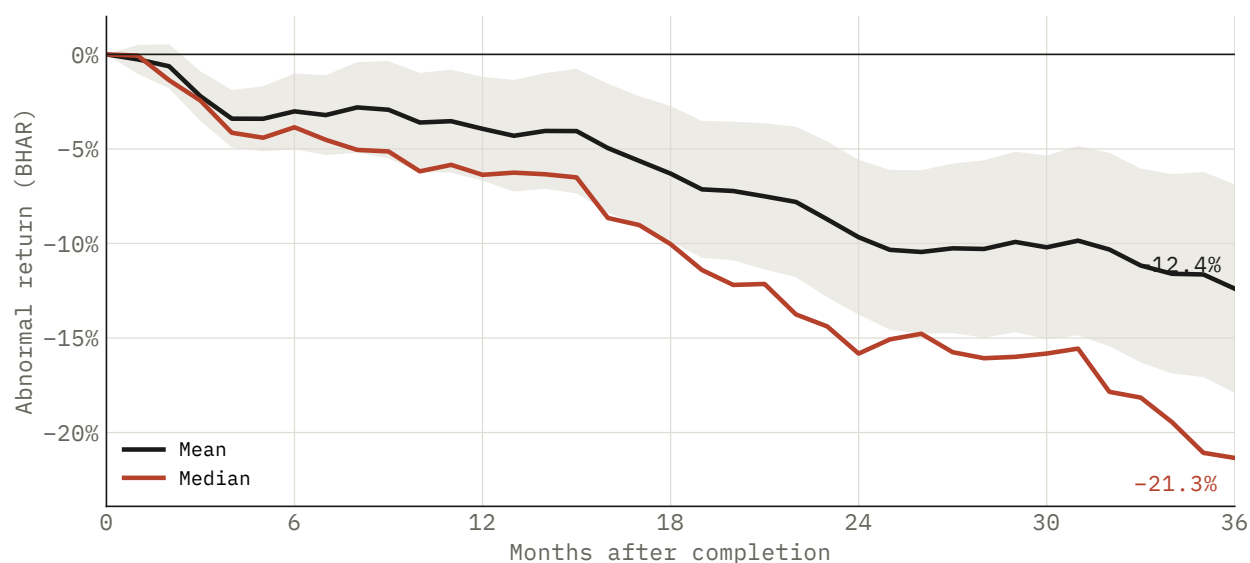
Differences between kinds of deal are shown as sorts on single traits (Table 6) and in a regression of the winsorized 36-month BHAR on all traits together, with industry dummies and standard errors clustered by completion year (Table 7). A placebo repeats the BHAR calculation on the same acquirers in three-year windows without a deal, as a check on whether the shortfall belongs to the deals or to the firms.

5 Results

5.1 Event time: the acquirer falls behind

Table 3 and Figure 2 show the central result. Twelve months after completion the average acquirer has returned 3.9% less than its benchmark; after 24 months the gap is 9.7% and after 36 months 12.4%. For the 641 deals with a full five-year window (completed by March 2020) the five-year gap is 18.6%. Every horizon is significant with the skewness-adjusted and bootstrapped statistics, and the median is consistently further below zero than the mean: the typical acquirer does worse than the average one, and a minority of very successful deals pulls the mean up. Only 37% of acquirers beat their benchmark over three years. The pattern is also persistent: there is no early drop followed by recovery, but a steady drift that is as steep in the second year as in the first.

FIGURE 2 MEAN AND MEDIAN BHAR, MONTHS AFTER COMPLETION



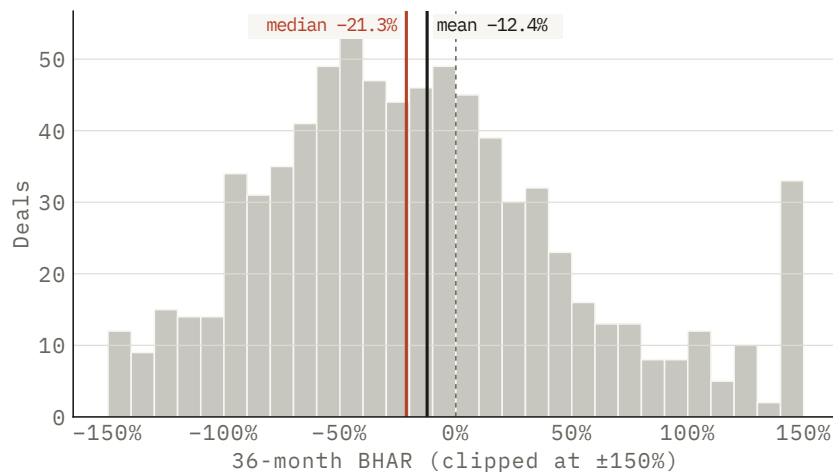
All 783 deals with a full 36-month window. Shaded: 95% confidence interval of the mean. Benchmark: value-weighted size and book-to-market portfolio.

TABLE 3 BUY-AND-HOLD ABNORMAL RETURNS AFTER COMPLETION

HORIZON / BENCHMARK	DEALS	MEAN	MEDIAN	BEAT	T	T SKEW- ADJ.	BOOT. P	WILCOXON P
SIZE AND BOOK-TO-MARKET PORTFOLIOS, VALUE-WEIGHTED (BASE CASE)								
12 months	783	-3.9%	-6.4%	43%	-2.81	-2.76	0.006	0.000
24 months	783	-9.7%	-15.8%	38%	-4.63	-4.52	0.000	0.000
36 months	783	-12.4%	-21.3%	37%	-4.40	-4.11	0.000	0.000
60 months	641	-18.6%	-38.6%	32%	-3.26	-2.36	0.042	0.000
ALTERNATIVE BENCHMARKS, 36 MONTHS								
Equal-weighted size and book-to-market portfolios	783	-11.0%	-19.8%	39%	-3.86	-3.64	0.000	0.000
Value-weighted market portfolio (Fama-French Mkt)	783	-18.0%	-28.9%	33%	-6.35	-5.73	0.000	0.000

Beat: share of deals with a positive BHAR. t skew-adj.: Johnson skewness-adjusted t-statistic; Boot. p: two-sided p-value from 2,000 bootstrap resamples of the skewness-adjusted statistic (Lyon, Barber and Tsai, 1999); Wilcoxon p: signed-rank test on the median. Returns include dividends. Sixty-month windows require completion by March 2020.

The result does not depend on the benchmark weighting: equal-weighted reference portfolios give -11.0% at 36 months. Against the market portfolio the shortfall is larger, -18.0%, because acquirers are smaller than the market average and small stocks lagged large ones over the period. Figure 3 shows the whole distribution. It is wide, with an interquartile range of well over 50 percentage points, which is why single deal stories prove little in either direction.

FIGURE 3 DISTRIBUTION OF 36-MONTH BHARS

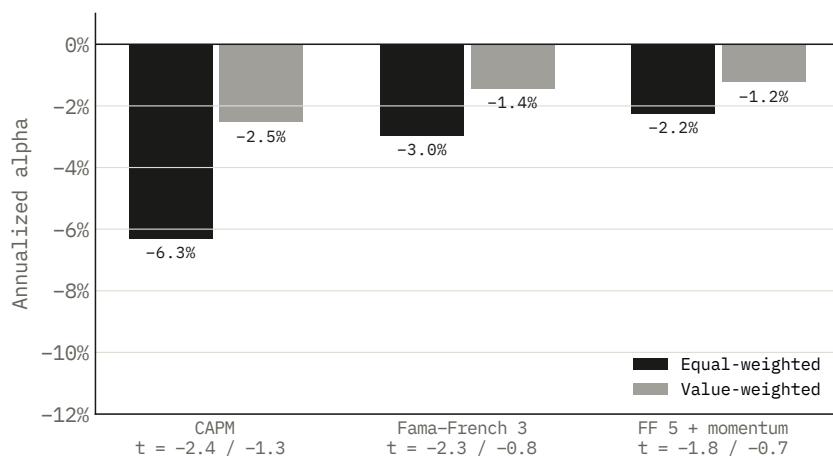
Values beyond $\pm 150\%$ are shown in the end bins. Vertical lines: mean and median.

5.2 Calendar time: much of the gap is risk and style

Table 4 and Figure 4 report the calendar-time regressions. With the CAPM, the equal-weighted acquirer portfolio has an alpha of -0.54% a month ($t = -2.45$), roughly 6% a year. Adding the size and value factors cuts it to -0.25% ($t = -2.32$), and the full five-factor plus momentum model to -0.19% ($t = -1.80$), significant only at 10%. Value-weighted, the five-factor alpha is -0.10% a month and not statistically significant.

The loadings explain the gap. The equal-weighted portfolio has a market beta of 1.14, a strong small-cap tilt (SMB 0.78) and a negative momentum loading (−0.13). A size and book-to-market portfolio matches the first two only roughly and ignores the third. Over a period in which high-beta, small and low-momentum stocks often lagged, those exposures alone produce a negative BHAR. This is the same pattern Mitchell and Stafford (2000) found for earlier decades: the event-time shortfall is real as a description of what shareholders earned, but most of it is explained by the kind of stock an acquirer is, not by something the acquisition did.

FIGURE 4 CALENDAR-TIME ALPHAS, ANNUALISED



Portfolio of all acquirers within 36 months of completion, monthly rebalanced. Bar labels: annualised alpha. Under each model: Newey-West t-statistics, equal-weighted / value-weighted.

TABLE 4 CALENDAR-TIME PORTFOLIO REGRESSIONS

MODEL	ALPHA / MONTH	T	ALPHA / YEAR	MKT	SMB	HML	RMW	CMA	MOM
EQUAL-WEIGHTED PORTFOLIO OF ACQUIRERS WITHIN 36 MONTHS OF COMPLETION									
CAPM	−0.54%**	−2.45	−6.3%	1.34					
Fama-French 3	−0.25%**	−2.32	−3.0%	1.17	0.82	0.17			
FF5 + momentum	−0.19%*	−1.80	−2.2%	1.14	0.78	0.10	−0.04	0.10	−0.13
VALUE-WEIGHTED PORTFOLIO OF ACQUIRERS WITHIN 36 MONTHS OF COMPLETION									
CAPM	−0.21%	−1.28	−2.5%	1.01					
Fama-French 3	−0.12%	−0.80	−1.4%	0.96	0.23	0.16			
FF5 + momentum	−0.10%	−0.72	−1.2%	0.97	0.23	0.01	−0.03	0.31	−0.09
BY PAYMENT AND SHARE ISSUANCE, FF5 + MOMENTUM, EQUAL-WEIGHTED									
Cash deals	−0.10%	−0.82	−1.2%						
Mixed or stock-heavy deals	−0.58***	−3.18	−6.8%						
Stock minus cash (long-short)	−0.54***	−2.65	−6.3%						
Shares up 10% or more	−0.41%**	−2.36	−4.8%						
Shares up less than 10%	−0.16%	−1.37	−1.9%						
Difference	−0.30%*	−1.66	−3.5%						

Monthly excess returns of the portfolio of all acquirers that completed a deal in the previous 36 months, 153 months from March 2012 to November 2024, on average 181 firms; months with fewer than 10 firms are dropped. Newey-West t-statistics (three lags). * * * * *: significant at 10%, 5%, 1%. Long-short rows: difference of the two portfolios, only months where both exist.

The notable exception is payment. The portfolio of acquirers that paid partly or mostly in stock has a five-factor alpha of -0.58% a month ($t = -3.18$), about 7% a year; cash acquirers earn -0.10% ($t = -0.82$), and the long-short difference of -0.54% a month is significant at 1% . Share issuance measured directly from the cover-page share count points the same way (-0.41% for acquirers whose shares rose 10% or more, against -0.16%), but the difference is not significant, partly because share counts also move for reasons unrelated to the deal.

5.3 Robustness

Table 5 changes one choice at a time. Lowering the relative-size threshold to 5% or raising it to 25% , keeping overlapping deals, dropping every ticker match not confirmed by the public float, using only cash-flow and equity tags for deal value and winsorizing all give 36-month means between -13.3% and -11.9% , all significant; equal-weighted benchmark portfolios give -11.0% and the market portfolio -18.0% . The event-time result is not an artefact of a particular sample choice; the question raised in section 5.2 is about what it measures, not whether it is there.

TABLE 5 ROBUSTNESS OF THE 36-MONTH BHAR

SPECIFICATION	DEALS	MEAN	MEDIAN	T SKEW-ADJ.
Base: size/book-to-market portfolios, value-weighted	783	-12.4%	-21.3%	-4.11
Equal-weighted size/book-to-market portfolios	783	-11.0%	-19.8%	-3.64
Market portfolio	783	-18.0%	-28.9%	-5.73
Deals of 5% or more of acquirer value	925	-12.7%	-21.1%	-4.70
Deals of 25% or more	444	-13.3%	-21.8%	-3.12
All deals, overlaps included	945	-11.9%	-21.3%	-4.13
Price matched to float only (drop name-only matches)	781	-12.4%	-21.3%	-4.10
Deal value from cash-flow and equity tags only	768	-12.3%	-21.2%	-4.02
Winsorized at 1% and 99%	783	-12.6%	-21.3%	-4.40
PLACEBO: THE SAME ACQUIRERS, NO DEAL				
Quiet three-year window after the deal (average of 50 draws)	203	+6.2%	-14.4%	0.78

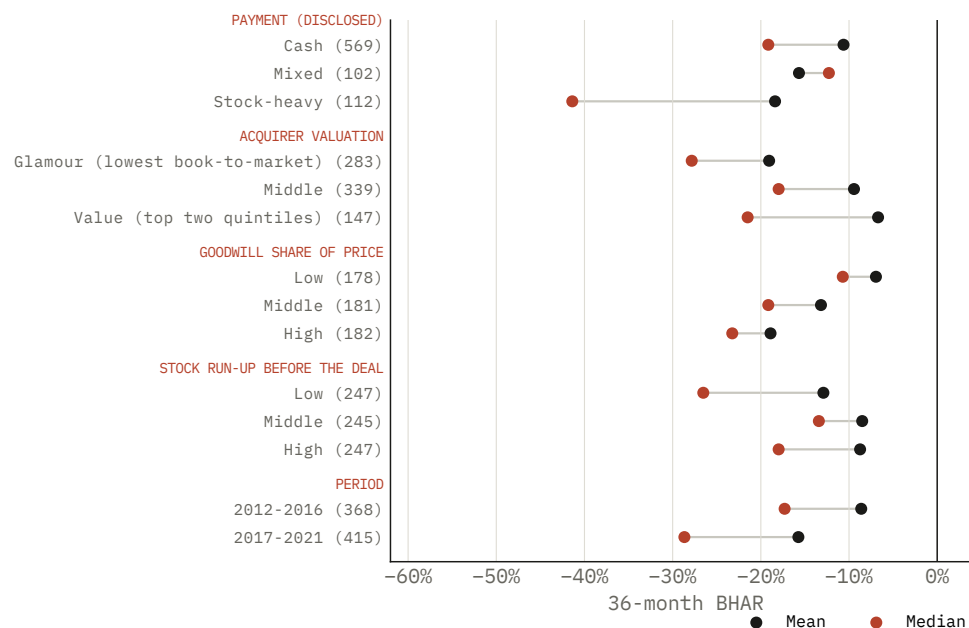
Each row changes one choice relative to the base case. The placebo draws, for each acquirer, a random start month after its first sample deal with no \$100m+ acquisition within three years either side, re-assigns the size and book-to-market portfolio at that date and computes the same BHAR.

The placebo is informative in a different way. The same acquirers, in three-year windows after their deal with no other large acquisition, earn a mean BHAR of $+6.2\%$ against the same kind of benchmark, with a median of -14.4% . The mean difference between deal and no-deal windows is large, which points to the deals. The negative placebo median is a reminder that long-horizon medians are negative for almost any group of stocks because of skewness, so the median BHAR after deals (-21.3%) should be read against that level rather than against zero. Placebo windows before the deal are deliberately excluded: they earn very high abnormal returns because only firms that later grew large enough to make a significant acquisition are in the sample, a look-ahead bias that would flatter the comparison.

5.4 Which deals do worse

Table 6 and Figure 5 sort the 36-month BHAR by deal and acquirer traits. Most sorts point in the direction the literature predicts. Glamour acquirers, in the lowest book-to-market quintile, average -19.1% against -6.7% for value acquirers, in line with Rau and Vermaelen (1998). Deals with the highest share of goodwill in the price, a rough measure of how much was paid above the target's identifiable assets, average -18.9% against -6.9% for the lowest tercile. Deals completed from 2017 to 2021 did worse than those from 2012 to 2016 (-15.7% against -8.6%). Stock-heavy deals have by far the lowest median (-41.4%), but their mean is close to that of cash deals, because a handful of stock deals did very well.

FIGURE 5 MEAN AND MEDIAN 36-MONTH BHAR BY GROUP



Black: mean; red: median. Number of deals in brackets.

TABLE 6 36-MONTH BHAR BY DEAL AND ACQUIRER CHARACTERISTICS

GROUP	DEALS	MEAN	MEDIAN	BEAT	T SKEW-ADJ.
PAYMENT (DISCLOSED)					
Cash	569	-10.6%	-19.2%	37%	-3.25
Mixed	102	-15.7%	-12.3%	43%	-1.73
Stock-heavy	112	-18.4%	-41.4%	30%	-1.77
NEW SHARES ISSUED AROUND THE DEAL					
Under 10%	553	-12.0%	-21.3%	38%	-3.61
10% or more	230	-13.4%	-21.8%	35%	-2.06
DEAL SIZE VS ACQUIRER					
Small (10% to 19.2%)	262	-16.8%	-23.7%	34%	-4.24
Medium	261	-7.9%	-19.3%	39%	-1.54
Large (38.8% or more)	260	-12.4%	-27.1%	37%	-2.00
ACQUIRER VALUATION					

GROUP	DEALS	MEAN	MEDIAN	BEAT	T SKEW-ADJ.
Glamour (lowest book-to-market)	283	-19.1%	-27.8%	31%	-3.41
Middle	339	-9.4%	-18.0%	40%	-2.35
Value (top two quintiles)	147	-6.7%	-21.5%	41%	-0.85
ACQUISITION HISTORY					
Occasional (0-1 prior)	574	-12.1%	-22.0%	37%	-3.36
Serial (2+ in 5 years)	209	-13.1%	-20.4%	37%	-2.40
STOCK RUN-UP BEFORE THE DEAL					
Low	247	-12.9%	-26.5%	37%	-2.35
Middle	245	-8.5%	-13.4%	39%	-1.67
High	247	-8.7%	-18.0%	38%	-1.69
GOODWILL SHARE OF PRICE					
Low	178	-6.9%	-10.7%	43%	-1.13
Middle	181	-13.2%	-19.2%	37%	-2.16
High	182	-18.9%	-23.2%	32%	-3.36
PERIOD					
2012-2016	368	-8.6%	-17.3%	40%	-2.21
2017-2021	415	-15.7%	-28.7%	34%	-3.46

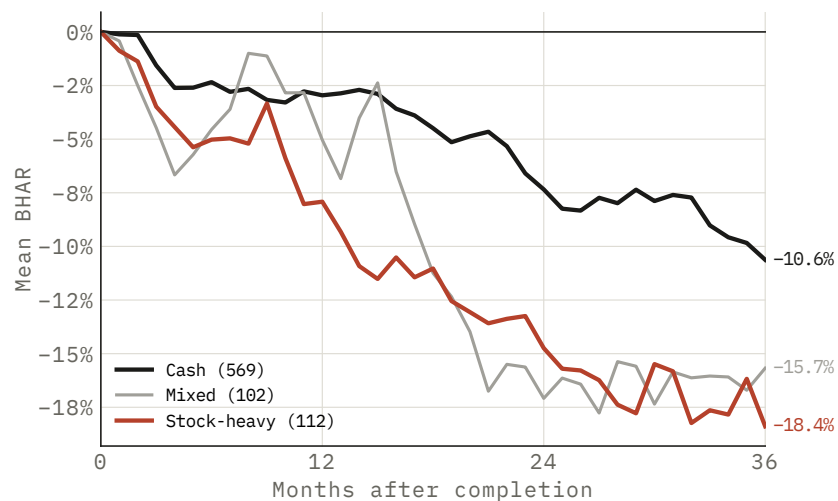
Payment: cash if stock is under 5% of the price paid, stock-heavy if at least 50%. New shares: change in cover-page shares outstanding from the last cover before completion to the first cover within nine months after. Run-up: BHAR over the 12 months before completion. Goodwill share: goodwill acquired divided by deal value, where disclosed. Size, run-up and goodwill groups are terciles; value acquirers are the top two book-to-market quintiles.

Table 7 puts the traits into one regression. None of them is individually significant once the others and industry are held fixed, and the regression explains about 4% of the variation. Two readings are possible and both are probably partly right: the traits overlap (glamour acquirers pay in stock, and stock deals carry more goodwill), and the dispersion of long-run returns is so wide that a sample of a few hundred deals cannot pin down effects of a few percentage points. For a deal team the practical conclusion is that these traits are warning signs to be weighed, not a scoring model.

TABLE 7 WHAT EXPLAINS THE DISPERSION: CROSS-SECTIONAL REGRESSION

VARIABLE	COEFFICIENT	T
Stock share of the price paid	-0.087	-0.54
Log (deal value / acquirer market value)	0.080	1.26
Log acquirer market value	-0.006	-0.23
Goodwill / deal value	-0.059	-0.44
Serial acquirer (0/1)	-0.025	-0.26
Run-up, 12 months before	-0.095	-0.94
Glamour acquirer (0/1)	-0.030	-0.29
Completed 2017 to 2021 (0/1)	-0.006	-0.08
Intercept	-0.312	-1.32
Industry dummies (first SIC digit)	yes	
Deals / R-squared	498	0.045

OLS of the 36-month BHAR, winsorized at 1% and 99%, on deal and acquirer traits. Standard errors clustered by completion year. The sample is smaller than the full 783 because goodwill acquired or the payment split is not tagged for every deal. With only ten completion years as clusters, the t-statistics are approximate.

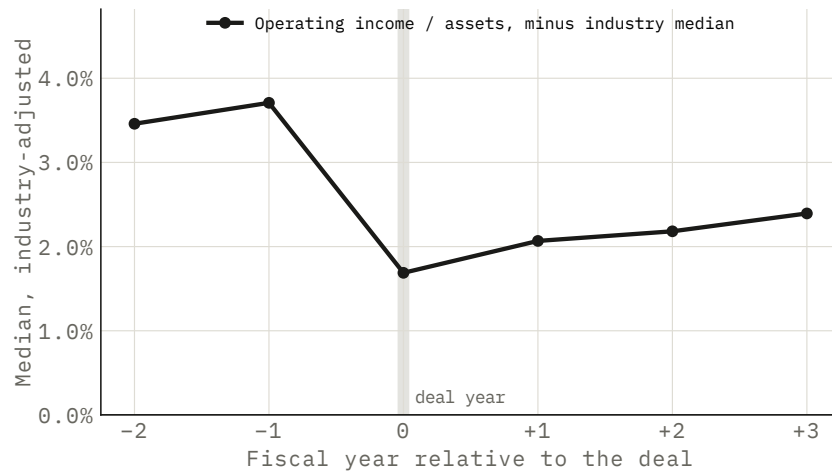
FIGURE 6 MEAN BHAR BY DISCLOSED PAYMENT

Deals with a full 36-month window, grouped by the stock share of the price paid.

5.5 Operating performance: no sign of synergies

Acquirers were more profitable than their industries before the deal: the median industry-adjusted return on assets was +3.8% in the year before completion (Table 8, Figure 7). It drops in the deal year, which is mechanical (assets jump at completion while the target contributes only part of a year's income) and recovers only partly: the median over years +1 to +3 is +2.1%. The Healy, Palepu and Ruback regression shows how much of that decline is simple persistence. The slope of 0.75 says that about three quarters of an acquirer's pre-deal advantage would be expected to survive three years in any case; the intercept of -0.44% ($t = -1.14$) is the abnormal change, and it is not statistically significant. The deals neither improved nor clearly damaged operating returns relative to what the acquirers' own history predicts. Cash-flow return on assets tells the same story, from a median of 9.9% before to 8.1% after.

FIGURE 7 INDUSTRY-ADJUSTED OPERATING RETURN ON ASSETS



Median operating income over average assets minus the two-digit SIC industry median, fiscal years relative to the deal year.

TABLE 8 OPERATING PERFORMANCE AROUND THE DEAL

FISCAL YEAR VS DEAL	FIRMS	ADJ. ROA, MEDIAN	ADJ. ROA, MEAN	CASH-FLOW ROA, MEDIAN
-2	700	+3.5%	+4.2%	9.7%
-1	724	+3.7%	+4.8%	9.8%
0 (deal year)	728	+1.7%	+3.0%	7.7%
+1	720	+2.1%	+2.7%	8.3%
+2	710	+2.2%	+2.9%	7.9%
+3	680	+2.4%	+4.5%	7.9%

HEALY, PALEPU AND RUBACK REGRESSION: $POST = A + B \times PRE$

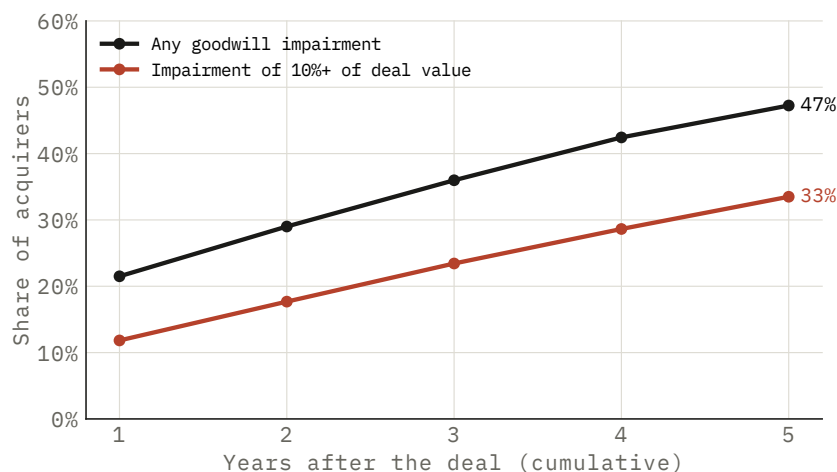
Intercept a (abnormal change)	696	-0.44%	t = -1.14	
Slope b (persistence)		0.749	t = 17.83	R ² 0.53

Adjusted ROA: operating income over average total assets minus the median of all SEC filers in the same two-digit SIC industry and fiscal year (industries with at least five firms with assets above \$10m). Means winsorized at 1% and 99%. Post: average of years +1 to +3 (at least two available); pre: year -1. Heteroskedasticity-robust t-statistics.

5.6 Goodwill impairments

Goodwill is where an overpayment eventually becomes visible in the accounts. Within one year of the deal, 22% of acquirers report a goodwill impairment; within five years 47% do, and 33% report cumulative impairments of at least a tenth of the deal value (Table 9, Figure 8). Not every impairment relates to the sample deal, and impairment tests are slow and discretionary, so these numbers are better read as a lower bound on how often the price paid proved too high than as a precise measure.

FIGURE 8 CUMULATIVE SHARE OF ACQUIRERS WITH A GOODWILL IMPAIRMENT



Deal year and following fiscal years, up to fiscal 2024.

TABLE 9 GOODWILL IMPAIRMENTS AFTER THE DEAL

HORIZON	ACQUIRERS	ANY IMPAIRMENT	AT LEAST 10% OF DEAL VALUE
Within 1 year	786	22%	12%
Within 2 years	786	29%	18%
Within 3 years	781	36%	23%
Within 4 years	681	42%	29%
Within 5 years	618	47%	33%

Cumulative share of acquirers reporting GoodwillImpairmentLoss in the deal year or later fiscal years, up to fiscal 2024. The impairment is not necessarily on the goodwill from this deal; the total is compared with the deal value.

6 Discussion

The results give a clear answer to the narrow question and a more careful one to the broad question. The narrow question is whether shareholders who held the acquirer through the three years after a significant deal did better than holders of comparable companies. On average they did not: they fell 12% behind, and nearly two in three fell behind. That is the number a board should keep in mind, because it is the experience of the acquirer's actual shareholders.

The broad question is whether the acquisitions themselves destroyed value. Here the evidence is weaker. Once returns are measured in calendar time and adjusted for the acquirers' factor exposures, the average shortfall shrinks to about 2% a year equal-weighted and is significant only at 10%; value-weighted it is not statistically significant. Acquirers are high-beta, smaller, low-momentum companies, and such companies lagged over the period whether or not they bought something. The operating evidence agrees: there is no abnormal improvement in margins, but no clear deterioration either. Put simply, the average large deal did not create value that shows up in returns or in the accounts, and it probably did not destroy much either; it left the acquirer carrying the risks it already had, plus the integration.

Implications for a deal team.

Start from the base rate. Two in three acquirers trail comparable stocks over three years. A deal case should explain why this one is different, in terms that can be checked against the acquirer's own history.

Treat stock consideration as a warning sign. Stock-financed acquirers lose about 7% a year against the factors. Whether that is because the deals are worse or because managers issue stock when it is expensive, the post-deal return is weak either way, and the seller accepting shares may know something about their value.

Scrutinise the goodwill. The deals with the most goodwill in the price did worst, and almost half of acquirers impaired goodwill within five years. The integration plan and the synergy case deserve more scrutiny than the strategic rationale, which is rarely what goes wrong.

Do not rely on a formula. No single trait explains the cross-section once the others are controlled for. Valuation, payment and goodwill are warning signs, not a model.

The results are consistent with the direction of the pre-2010 literature but with smaller magnitudes. The stock-cash split is similar in spirit to Loughran and Vijh (1997), though the cash deals here do not outperform. The fragility of the average result under calendar-time methods is what Fama (1998) and Mitchell and Stafford (2000) would predict. The weak value-weighted result is also consistent with Loughran and Ritter's (2000) point that value-weighted tests have low power; the largest acquirers in the sample did no worse than their factor exposures predicted.

7 Limits

- **Deal identification.** Deals come from acquisition spending in XBRL filings and 8-K Item 2.01 dates rather than a commercial database. A year with several deals becomes one event dated at the first completion, targets are not identified, and deals by companies that did not file XBRL or did not file an Item 2.01 are missing.
- **Payment.** The payment split relies on what filers tag. Some large mergers tag the stock component under custom elements, so stock deals are probably under-counted and a few are classified as cash. A share-count measure of issuance is reported alongside to limit the damage.
- **Prices.** 83% of candidate deals could be matched to a validated ticker; the rest are lost, which may favour firms that were easier to trace. The price and dividend data come from a community-maintained database; dividend amounts are split-adjusted for some securities and not for others, and amounts around large splits were rescaled.
- **Benchmark.** Size and book-to-market portfolios do not control for industry, momentum or profitability. The calendar-time regressions address the last two but not the first.
- **Placebo.** Quiet windows are not a random draw: they require three years without a large deal, which may favour firms that were doing well without acquiring. The comparison is suggestive, not a clean control.
- **Causality.** Long-run abnormal returns describe association. The counterfactual, the same firm without the deal, is not observed, and Savor and Lu (2009) show that for stock deals it may have been worse.
- **Point in time.** Prices, returns, filings and accounting periods end on 31 March 2025, and cover-page data are used only when they were public by then. Accounting values come from the SEC frames API, which reports each company's latest filed figure for a period, so a later restatement of a pre-2025 figure can appear. Five-year results cover deals completed by March 2020 only.
- **Missing first month.** 3 deals have no price in the first month after completion, a ticker change the data do not bridge, and are left out of the return tests.

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